SAMMAM MAHDI

sammam.mahdi@gmail.com +880 1822926571 Dhaka, Bangladesh
Google Scholar GitHub LinkedIn

Research Interests

3D computer vision; reliability and uncertainty of 3D foundation models; feed-forward 3D reconstruction and monocular metric 3D perception; trustworthy vision and media forensics.

Education

B.Sc. in Computer Science and Engineering *Graduated June 2026*
BRAC University, Dhaka, Bangladesh

CGPA: 3.92 / 4.00 | **Honors:** Highest Distinction, Vice-Chancellor's List

Relevant coursework: Artificial Intelligence (A), Computer Graphics (A), Introduction to Robotics (A+), Numerical Methods (A), Algorithms (A), Data Structures (A+), Operating Systems (A).
Mathematics: Calculus I (A) & II (A+), Linear Algebra & Fourier Analysis (A), Complex Variables & Laplace Transforms (A+), Statistics & Probability (A), Discrete Mathematics (A).

Edexcel International A-Levels (2021) & IGCSE (2019) — Average Grade A.

English proficiency: IELTS Academic 7.5 overall (October 2026)

Publications

Under review (not yet published)

[U1] **ReVar3R: Gauge-Aware Perturbation Uncertainty for Feed-Forward 3D Reconstruction. S. Mahdi**, F. B. Salim, R. B. Rabbani, A. N. Zereen. Under review at a top-tier ML conference; preprint *arXiv:2610.07883*, 2026. [arXiv]
Showed in closed form that run-to-run similarity-frame (Sim(3)) variation contaminates training-free perturbation uncertainty from frozen feed-forward 3D models, and that aligning runs before taking variance removes it. With no retraining and no labels, this lowers sparsification error (AUSE) below the models' built-in confidence in 11 of 18 backbone–dataset conditions (15 of 18 with held-out fusion), across VGGT, MASt3R and π^3 on six datasets.

[U2] **CAMDet: Camera-Agnostic Monocular Metric 3D Object Detection with Learned Intrinsic Feature Conditioning.** R. U. Karim, **S. Mahdi**, A. Samin, S. M. Tabassum, R. B. Rabbani, A. N. Zereen. Under review at *IEEE Transactions on Image Processing*, 2026.
A single-stage DINOv2-DETR detector conditioned on camera intrinsics (FiLM). On sensors held out from training, it beats a Cube R-CNN-style virtual-depth control by 6.3 AP3D@0.25 (SUN RGB-D), and one checkpoint runs unmodified from a desktop GPU to a Jetson Orin Nano.

Peer-reviewed

[J1] **Optimizing Stroke Recognition with MediaPipe and Machine Learning: An Explainable AI Approach for Facial Landmark Analysis.** R. U. Karim, **S. Mahdi**, A. Samin, A. N. Zereen, M. Abdullah-Al-Wadud, J. Uddin. *IEEE Access*, vol. 13, pp. 32636–32660, 2025. [DOI] [Code]
Built the XAI pipeline (228 facial landmarks) and the Multimodal Voting Classifier (RF + XGBoost + CatBoost) — 94.75% accuracy, 96.5% recall.

[C1] **Vision-based Hand Gesture Virtual Keyboard-Mouse Framework with Bilingual Next-word Prediction. S. Mahdi**, R. U. Karim, T. Sujat, S. M. Tabassum, A. Samin, A. N. Zereen. *23rd Intl. Joint Conf. on Computer Science and Software Engineering (JCSSE)*, Bangkok, 2026. [DOI]

[C2] **Improved Photoplethysmography-Based Four-Stage Sleep Classification with Explainable AI-Driven Machine Learning.** T. Ferdous, R. U. Karim, A. Samin, **S. Mahdi**, A. N. Zereen. *IEEE ICEACE*, 2024. [DOI]

[C3] **Machine Learning Approaches in Photoplethysmography-Based Sleep Stage Classification.** T. Ferdous, R. U. Karim, A. Samin, **S. Mahdi**, H. Tasnim, A. N. Zereen. *IEEE ICEACE*, 2024. [DOI]

Research Experience

Research Assistant — Dr. Aniqua Nusrat Zereen's group *2024 – Present*
CSE, BRAC University | undergraduate researcher from 2024; Undergraduate Research Assistant since Nov 2025

- **ReVar3R** [U1], first author: conceived the problem, derived the closed-form gauge-contamination result, designed the estimator, and ran all experiments across VGGT, MASt3R and π^3.
- **CAMDet** [U2], second author: co-designed the detection pipeline with the first author and ran the experiments and ablations.
- **DRISTEE** (undergraduate thesis, first author, grade A+, RSGI-funded): a six-module assistive agent for blind and low-vision users driven by one RGB camera — CAMDet metric 3D detections fused with MASt3R-SLAM into a metric traversability map for path planning with spoken guidance, plus VLM scene description, face recognition, OCR, and voice interaction.
- **Cervical Cancer Screening** group: contributed to and supervised an edge-deployable cytology classifier; guided implementation and evaluation.
- Co-authored journal and conference papers on explainable facial-landmark analysis and physiological-signal classification ([J1], [C2], [C3]).

Ongoing Research

ICEA — Invariant-Counterfactual Evidence Accumulation *In progress*
Deepfake detection from violations of physical and human invariants rather than generator-specific artifacts, aiming to generalize to unseen generators; a dual-encoder backbone with a counterfactual violation classifier for forensic attribution.

Edge-Native Multimodal Agent for Assistive Perception *In progress*
An agentic system (planner/router, tool use, vector memory, multi-LLM/VLM integration, retrieval) with a structured evaluation harness and anti-fabrication guardrails; reduced cloud vision calls by over 95%.

Teaching Experience

Teaching Assistant — Data Structures & Algorithms *Jan 2025 – May 2026*
CSE Department, BRAC University

- Led weekly recitations for **100+ students** on complexity analysis, graph algorithms, and dynamic programming; designed problem sets, evaluated assessments, and held one-on-one consultations.
- Mentored a team of undergraduate researchers — reviewed weekly progress, guided experiments, and coordinated toward publication.

Robotics & Engineering

BRACU Mongol Tori — Mars Rover Team *2023 – 2025*
*University Rover Challenge (URC), Utah, USA | team ranked **8th globally** at URC 2025*

- **AI/ML core team** (2024–2025): trained YOLOv8 detectors for real-time object perception under outdoor variability; optimized them for on-device inference on NVIDIA Jetson.
- **Control team** (2023–2024): developed the rover's ROS control systems — teleoperation and motion control for manual and autonomous navigation.

Metadata Journaling File System Code
C, Operating Systems. Crash-consistent journaling file system with an `mkfs` image builder and a 16-block write-ahead journal (DATA/COMMIT records) with crash-recovery replay.

SkillWise — AI-Powered E-Learning & Anti-Cheat Code
Computer Vision + MERN. E-learning platform with real-time MediaPipe proctoring and CV-based course recommendation. **3rd Runner-up**, Software Engineering Showcase (200+ teams).

Honors & Awards

- **Highest Distinction** (graduation) and **Vice-Chancellor's List** (multiple semesters), BRAC University.
- **Research Seed Grant Initiative (RSGI)** — competitive research grant, BRAC University, for the DRISTEE thesis project.
- **Globally Ranked 8th, University Rover Challenge 2025** — The Mars Society (Team BRACU Mongol Tori).
- **3rd Runner-up, Software Engineering Showcase** — out of 200+ teams (SkillWise).
- **HULT Prize Semi-Finalist** — international social-entrepreneurship competition.
- **The Daily Star Award** — national recognition for outstanding O/A-Level results, Bangladesh.
- **Edexcel High Achievers' Award** — for top performance in Edexcel International examinations.
- **The Duke of Edinburgh's International Award — Bronze**

Technical Skills

3D Vision:	Feed-forward 3D foundation models (VGGT, MASt3R, π^3), similarity (Sim(3)) registration, monocular metric depth, 3D object detection
Uncertainty & Evaluation:	Perturbation ensembles, sparsification metrics (AUSE), isotonic calibration, paired statistical testing
ML & Vision:	PyTorch, TensorFlow, Hugging Face Transformers, DINOv2, scikit-learn, XGBoost, OpenCV, MediaPipe, YOLOv5/v8, SHAP, Grad-CAM
Generative AI & Agents:	LLMs, VLMs, RAG pipelines, agent tool use, evaluation harnesses
Systems & Robotics:	Linux, Git, Docker, ROS, NVIDIA Jetson (edge inference)
Languages:	Python, C, C++, Java, JavaScript, SQL

References

Dr. Aniqua Nusrat Zereen	aniqua@bracu.ac.bd — Assistant Professor, CSE, BRAC University (Research & Thesis Supervisor)
Dr. Farig Sadeque	farig.sadeque@bracu.ac.bd — Assistant Professor, CSE, BRAC University
Dr. Md. Khalilur Rahman	khalilur@bracu.ac.bd — Associate Professor, CSE, BRAC University